

Lesson 8

Greatest common divisor

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Objectives

At the end of this session you should be able to

- Define greatest common divisor (gcd) and least common multiplier (lcm)
- Prove the existence of integers x and y such that $\gcd(a, b) = ax + by$

8.1 Introduction

Greatest common divisor and least common multiple are not new concepts to you. You are familiar with these concepts from your school days. Perhaps you encountered with these concepts first, when you were preparing for the grade five scholarship exam. Those concepts have been introduced to you without those names, but as real world applications.

In this session we introduced these concepts formally and we develop and set up these concepts in an abstract frame work.

8.2 Greatest Common Divisor

We start with the following real world problems.

- Nimali wants to give some friends bags with chocolates and toffees in them. If she has 20 chocolates and 15 toffees to split equally among the bags, and wants no chocolates or toffees left over, what is the greatest number of bags Nimali can make?

- Nimal has two pieces of Bamboo, one 15 feet long and the other 18 feet long. If he wants to cut them up to produce many pieces of Bamboo that are all of the same length, with no Bamboo left over, what is the greatest length, in feet, that he can make them?

The answers to these two questions rely on the concept of greatest common divisor.

Highest common factor is another name to greatest common divisor. Now let us define greatest common divisor formally.

8.2.1 Definition

Let a and b be given integers, with at least one of them different from zero. The greatest common divisor of a and b , denoted by $\gcd(a, b)$ is the positive integer d satisfying the following;

- (i) $d|a$ and $d|b$
- (ii) If $c|a$ and $c|b$, then $c \leq d$.

8.2.2 Example

Find the greatest common divisor of -12 and 30.

Answer:

The positive divisors of -12 are 1, 2, 3, 4, 6, and 12. The positive divisors of 30 are 1, 2, 3, 5, 6, 10, 15, 30. Hence, the positive common divisors of -12 and 30 are 1, 2, 3, 6. Since 6 is the largest of these integers, it follows that $\gcd(-12, 30) = 6$.

8.2.3 Example

Find the greatest common divisor of 9 and 10.

Answer:

The positive divisors of 9 are 1, 3, and 9. The positive divisors of 10 are 1, 2, 5, and 10. Hence, the only positive common divisors of 9 and 10 is 1. Hence $\gcd(9, 10) = 1$.

8.2.4 Example

If a is a non zero integer, then find the greatest common divisor of a and $2a$.

Answer:

$|a|$ is the largest positive divisor of a . Also $|a|$ is a positive divisor of $2a$. Also an integer larger than $|a|$ is not a divisor of $|a|$. Hence $|a|$ is the largest common divisor of a and $2a$. Therefore $\gcd(a, 2a) = |a|$.

The next theorem states that $\gcd(a, b)$ can be represented as a linear combination of a and b . That is we can always find some integers x and y such that $\gcd(a, b) = ax + by$. Notice that again, well ordering principle plays a major role in the proof of the theorem.

8.2.5 Theorem

Given integers a and b , not both of which are zero, there exist integers x and y such that $\gcd(a, b) = ax + by$.

Proof:

Consider the set S of all positive linear combination of a and b :

$$S = \{au + bv \mid au + bv > 0; u, v \text{ integers}\}$$

Clearly S is a subset of integers. Let $u = a$ and $v = b$, then $au + bv = a^2 + b^2 > 0$. (since at least one of a or b is non zero)

Hence $a^2 + b^2 \in S$.

This shows the set S is non empty sub set of nonnegative integers. Hence by the well ordering principle, S must contain a least element. Let it be d .

Also by the definition of S , there exist integers x and y such that $d = ax + by$.

Now let's prove that $d = \gcd(a, b)$.

First we prove that d is a common divisor of a and b .

By the division algorithm $a = qd + r$ where $0 \leq r < d$.

Hence $r = a - qd = a - q(ax + by) = a(1 - qx) + b(-qy)$.

Now if $r > 0$ then $r \in S$.

But since $r < d$ this contradicts with the least element of S is d .

Therefore $r = 0$ and hence $a = qd$ or equivalently $d|a$.

By the similar argument $d|b$.

Hence d is a common divisor of a and b .

Now let's prove that d is the greatest common divisor of a and b .

Let c be an arbitrary positive common divisor of a and b .

Since $c|a$ and $c|b$, $c|ax + by$. This implies $c|d$.

Hence $c = |c| \leq |d| = d$.

This shows that d is the greatest common divisor of a and b . ■

Following definition is very common in the field of Number Theory.

8.2.6 Definition

Two integers a and b , not both of which are zero, said to be relatively prime whenever $\gcd(a, b) = 1$.

Combining this definition with theorem1, we can state the following theorem.

8.2.7 Theorem

Let a and b be integers, not both zero. Then a and b are relatively prime if and only if there exist integers x and y such that $ax + by = 1$.

Proof:

Suppose a and b are relatively prime so that $\gcd(a, b) = 1$.

Then Theorem3.2.5 guarantees the existence of integers x and y satisfying $ax + by = 1$.

As for the converse, suppose $ax + by = 1$ for some integers x and y .

Let $\gcd(a, b) = d$.

This yields $d|ax + by$ hence $d|1$.

Since d is positive integer, $d = 1$. ■

The following two theorems are very useful in later sessions

8.2.8 Theorem

If $a|c$ and $b|c$, with $\gcd(a, b) = 1$ then $ab|c$.

Proof:-

Since $a|c$ and $b|c$, there exist r and s such that $c = ar = bs$.

Now since $\gcd(a, b) = 1$, there exist integers x and y such that $ax + by = 1$.

Hence $c = c \cdot 1 = c(ax + by) = acx + bcy$. By substituting $c = ar = bs$ on right hand side, $c = a(bs)x + b(ar)y = ab(sx + ry)$. Hence $ab|c$.

8.2.9 Theorem (Euclid's Lemma)

If $a|bc$ with $\gcd(a, b) = 1$ then $a|c$.

Proof:-

Since $\gcd(a, b) = 1$, there exist integers x and y such that $ax + by = 1$.

Multiplying by c we obtained $c = acx + bcy$. Now since $a|bc$ and $a|ac$, we obtained $a|c$.

8.2.10 Example

Show that the numbers $6k+5$ and $5k+4$ are relatively prime for every integer k .

Answer:

$5(6k+5) - 6(5k+4) = 1$. Hence by Theorem 3.2.7, $6k+5$ and $5k+4$ are relatively prime.

Activity

Let a, b and c be non zero integers Show that:

$$(i) \gcd(a, b) = \gcd(a, -b) = \gcd(-a, b) = \gcd(-a, -b)$$

(ii) $\gcd(ac, bc) = c\gcd(a, b)$ if $c > 0$

(iii) if p is prime, then $\sqrt{p}$ is irrational

(iv) $\sqrt{35}$ is irrational

(v) $\sqrt{\frac{5}{7}}$ is irrational

(vi) Find integers x and y such that $13x + 21y = -5$